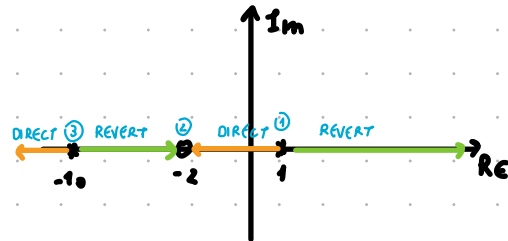


DRAW THE ROOT LOCUS OF A FUNCTION

GIVEN $G(s) = \frac{s+2}{(s+1)(s-1)}$



2) FIND POLES AND ZEROS

$z = \{-2\}$

$p = \{-1, 1\}$

3) DRAW THE GRAPH

- X-AXIS Re
- Y-AXIS Im
- ADD ON THE X-AXIS THE POLES ("x")
- ADD ON THE X-AXIS THE ZEROS ("o")

4) DRAW ARROWS

- FROM POLES $\rightarrow$ TO ZEROS
- MAKE ARROW TO $-\infty$ AND $+\infty$

5) CLASSIFY DIRECT AND REVERT LOCUS

- COUNT FROM RIGHT TO LEFT ($\leftarrow$) POLES AND ZEROS ①②③
- **DIRECT**: ARROW LEFT OF AN ODD NUMBER ①③
- **REVERT**: OTHERWISE (RIGHT OF AN ODD NUMBER) ②

6) COUNT THE ASYMPTOTES ν

- $\nu = \#P - \#Z$
- IF $\nu = 0 \rightarrow$ NO ASYMPTOT
 - IF $\nu = 1 \rightarrow \begin{cases} 1 \text{ ASYMPTOT FOR DIRECT} \\ 1 \text{ ASYMPTOT FOR REVERT} \end{cases}$
 - IF $\nu = 2 \rightarrow \begin{cases} 2 \dots \\ 2 \dots \end{cases}$

7) COMPUTE x_a (CENTRE OF ASYMPTOTS) ONLY FOR $\nu \geq 1$

$$x_a = \frac{\sum z_i - \sum p_i}{\nu} = \frac{\triangle \text{ OPPOSITE SIGN OF POLES AND ZEROS } 2 - (-1 - 1)}{2 - 1} = -7$$

8) CALCULATE ANGLE OF ASYMPTOTS ONLY FOR $\nu \geq 1$

- IF $\nu = \#P - \#Z = 1 \rightarrow \begin{cases} 180^\circ \text{ FOR DIRECT LOCUS} \\ 0^\circ \text{ FOR REVERT LOCUS} \end{cases}$ "ASYMPTOTS ARE ON THE REAL AXIS"

(RARE)

- IF $\nu = 2$ OR MORE $\rightarrow \begin{cases} \varphi_0 = \frac{(2k+1)(180^\circ)}{\nu} \quad (360^\circ k + 180^\circ)/\nu \\ \varphi_R = \frac{2k \cdot 180^\circ}{\nu} = 360^\circ k/\nu \end{cases} \quad k = 0, \dots, \nu-1$

9) FIND GAIN TO PLACE A POLE (ROOT) IN $\angle V$ (CORE)

- FIND THE VALUE ON THE X-AXIS
- IF THE ARROW IN THAT POINT IS DIRECT $\rightarrow$ "I EXPECT A POSITIVE GAIN"
- IF THE ARROW IN THAT POINT IS REVERT $\rightarrow$ "I EXPECT A NEGATIVE GAIN"

$P = \pm \frac{\prod m_i}{\prod \lambda_i} = - \frac{\text{EX: } -3}{7 \cdot 4}$

+ FOR DIRECT GAIN
- FOR REVERT GAIN

- m_i = DISTANCE OF POLES FROM THE POINT CONSIDERED

- λ_i = DISTANCE OF ZEROS FROM THE POINT CONSIDERED

DISTANCE ARE ALWAYS POSITIVE

SET OPEN LOOP CONTROLLER THAT UNDER NOMINAL CONDITION GUARANTEES...

$$G(s) = \frac{10(s+1)}{25s+1}$$

CONSTRAINTS

- SETTLING TIME 5% IN $[10; 20]$ SEC. $\longleftrightarrow 10 \leq T_{AS} \leq 20$ SECONDS
- RELATIVE ERROR LOWER THAN 5%. $\longleftrightarrow e_m < 10\%$

IDENTIFY THE SYSTEM

- P = POLES $\rightarrow$ NUMBERS THAT MAKES DENOMINATOR = 0
- Z = NUMBER THAT MAKE NUM. = 0
- # OF POLES $\rightarrow$ ORDER
- ORDER N = ORDER D $\rightarrow$ PROPER
- ORDER N $\neq$ ORDER D $\rightarrow$ STRICTLY PROPER

OA CHECK $G^*(s)$ IS STABLE

- IF $G^*(s) \exists p > 0 \rightarrow$ UNSTABLE SOLVE STABILITY
- IF $G^*(s) \forall p < 0 \rightarrow$ STABLE $\checkmark$ GO ON

OB STUDY OVERSHOOT (IF THERE IS A CONSTRAINT)

- NO OVERSHOOT $\rightarrow$ IF $G(s)$ IS 1st ORDER (ONLY ONE P) IS OK $\checkmark$
- MAX $\xi = 0.2 \rightarrow \xi = \frac{\gamma_m}{100} \rightarrow \gamma_m > 100 \cdot \xi \rightarrow \dots \rightarrow \gamma_m > 20^\circ$ PHASE MARGIN

POLE PLACEMENT SKETCH (OPEN LOOP) $G_o =$ DESIRED BEHAVIOUR = $\frac{N}{Ts+1}$

1 FIND N FROM ERROR%. IF "NO ERROR AT REGIME" = 0%.

$$|N-1| < 5\% \rightarrow -5\% < N-1 < 5\% \rightarrow 0.95 < N < 1.05 \rightarrow N=1$$

2 FIND T FROM SETTLING TIME %.

$$10 < T_{AS} < 20 \rightarrow 10 < 3T < 20 \rightarrow 10/3 < T < 20/3 \rightarrow T=5s$$

3 OBTAIN $G_o(s)$ FROM N AND T

$$G_o(s) = \frac{N}{Ts+1} = \frac{1}{5s+1}$$

4 OBTAIN $C(s)$ FROM $G_o(s)$ AND $G(s)$

$$C(s) = \frac{G_o(s)}{G(s)} = \frac{1}{5s+1} \cdot \frac{25s+1}{10(s+1)}$$

5 ANALYZE

- **FEASIBILITY** $\rightarrow$
 - $\deg(N) \leq \deg(D)$ FEASIBLE $\checkmark$
 - $\deg(N) > \deg(D)$ NOT FEASIBLE $\times$
- **NO UNSTABLE ZERO/POLE CANCELLATION** $\rightarrow$
 - NO SIMPLIFICATION OF POLES/ZEROS $\checkmark$
 - SIMPLIFICATION OF POLES/ZEROS $\times$ RARE
- **UNSTABLE POLES** $\rightarrow$
 - STABLE: ALL POLES (P) ARE NEGATIVE $\checkmark$
 - UNSTABLE: AT LEAST ONE POLE (P) > 0 $\times$

SOLUTION ADD A NEW POLE A DECADE AFTER THE BIGGEST FREQUENCY.

- TAKE BIGGEST FREQUENCY (EX: $s+100$) IN ABSOLUTE VALUE
- COMPUTE T ($T=1/\xi=1/100$)
- COMPUTE ξ ($\xi=1/T=100$ Hz)
- ADD A DECADE (10^2 Hz $\rightarrow 10^3$ Hz)
- ADD NEW POLE TO $G(s)$

$$C = \frac{1}{5s+1} \frac{s(s+100)^2}{s+10} \text{ NOT FEASIBLE } \times$$

$$C = \frac{1}{5s+1} \frac{s(s+100)^2}{s+10} \frac{1}{5/10^3+1} \text{ FEASIBLE } \checkmark$$

CLOSED LOOP

SOLUTION FOR UNSTABLE POLES

OPTION 1 - ROOT LOCUS

1 CHOOSE A GAIN S.T. THERE IS NO OVERSHOOT

- IF 1st ORDER $\rightarrow$ BY DEFINITION NO OVERSHOOT

2 DRAW ROOT LOCUS

- SEE ABOVE HOW TO DO IT

3 COMPUTE ν

$$\nu = \#P - \#Z$$

- IF $\nu=0$ NO ASYMPTOTES (PAC-MAN)
- IF $\nu=1$ 1 ASYMPTOTE (1 DIRECT AND 1 REVERT)

4 SOLVE CONSTRAINT FOR P AND PICK P

$$1+P G(s)=0$$

REMEMBER: $s < 0$ $\leftarrow$ NO POLE SHOULD BE GREATER THAN 0

$\dots \rightarrow -1 < P < 10$ $\leftarrow$ A GOOD SOLUTION IS $P=1$

5 DESIGN THE CONTROLLER C

$$G_o = P G(s) = 1 \cdot G(s)$$

$$C = \frac{G_o}{1+G_o}$$

6 FINAL CHECK

- STABLE $\rightarrow \forall p < 0$ $\checkmark$
- FEASIBLE $\rightarrow \deg(N) \leq \deg(D)$ $\checkmark$

OPTION 2 - BODE DIAGRAM

1 CHECK GAIN

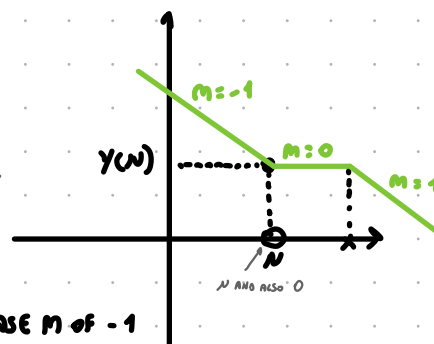
- IF $\eta \geq 1$ THEN $e_m=0$
- REMEMBER: $\eta = \# \text{ZEROS IN P} - \# \text{ZEROS IN Z}$

2 FIND N (GAIN) AND $\gamma(N)$

- $N = G(0)$ NEED TO EXCLUDE THE ZEROS IN P AND Z
- $\gamma(N) = 20 \log_{10}(|N|)$

3 CREATE THE GRAPH

- X-AXIS: POWER OF 10
- ADD POLES "x" AND ZEROS "o"
- LOCATE $(N; \gamma(N))$
- IF $\eta = +1$ START WITH $m = -\eta$ ANGULAR COEFFICIENT
- IF $\eta = 0$ START WITH $m = -\eta$
- EVERY TIME YOU MEET A POLE "x" DECREASE M OF -1
- EVERY TIME YOU MEET A ZERO "o" INCREASE M OF +1



4 CUT-OFF FREQUENCY ω_c

- FIND AT WHICH X THE GRAPH INTERCEPT $\gamma=0$ (EX: $\omega=10^{-4}$)
- TO FIND THE CUT-OFF FREQUENCY GO A STEP BACK IN THE X-AXIS (EX: $10^{-4} \rightarrow 10^{-2}$)

REMEMBER: • $j^2 = -1$

• $|a+ib| = \sqrt{a^2+b^2}$

• $\angle G(j\omega) = \text{ARG}(G(j\omega))$

• $\text{ARG}(A/B) = \text{ARG}(A) - \text{ARG}(B)$

• $\text{ARG}(a+ib) =$

- $b > 0 \rightarrow \tan^{-1}\left(\frac{b}{a}\right)$
- $a > 0; b < 0 \rightarrow \tan^{-1}\left(\frac{b}{a}\right) + 180^\circ$
- $a < 0; b < 0 \rightarrow \tan^{-1}\left(\frac{b}{a}\right) - 180^\circ$
- $a > 0; b = 0 \rightarrow +90^\circ$
- $a < 0; b = 0 \rightarrow -90^\circ$

5 CALCULATE THE PHASE-MARGIN

• WE REPLACE s WITH $w_c \cdot j$ CUT OFF FREQUENCY $\rightarrow G(s) \rightarrow G(w_c j)$

EX: $\frac{s-1}{s(s+10)} \rightarrow \frac{10^{-2}j-1}{10^{-2}j(10^{-2}j+10)}$

$\angle G(10^{-2}j) = \text{ARG}(G(10^{-2}j)) = (10^{-2}j-1) - \text{ARG}(10^{-2}j) - \text{ARG}(10^{-2}j+10)$

$= \tan^{-1}(10^{-2}/-1) - \tan^{-1}(10^2/0) - \tan^{-1}(10^{-2}/10) = (0+180) - (0+90) - 0 = 90^\circ$

FINALLY: $\varphi_m = 180 - |\angle G(10^{-2}j)| = 90^\circ \leftarrow \text{MUST BE GREATER THAN } \varphi_m \text{ (PHASE MARGIN)}$

EXERCISE 3 DESIGN CONTROLLER SUCH THAT CONSTRAINTS...

$$G(s) = \frac{100}{s(10s+1)(s+100)}$$

- OVERSHOOT LOWER THAN 15% $\rightarrow \xi < 15\%$
- ERROR LOWER THAN 5% $\rightarrow e_{ss} < 5\%$
- SETTING TIME AT MOST 10 SEC $\rightarrow T_s \leq 10 \text{ SEC}$
- NOISE MEASUREMENT ATTENUATION OF AT LEAST 20 dB?

• BUILD THE REGULATOR R_1 (STATIC PART)

- IF $\gamma \geq 1$ OF $G(s)$ $\rightarrow R_1 = N$ $\checkmark$
- IF $\gamma < 1$ OF $G(s)$ $\rightarrow$ CHOOSE A POWER OF s OF $R = N/s$ SUCH THAT $G(s) \cdot R$ HAVE $\gamma = 1$
- COMBINE R_1 AND $G(s)$ $\rightarrow R_1 \cdot G(s) = \frac{100N}{s(10s+1)(s+100)}$

• COMPUTE PHASE MARGIN γ_m FROM OVERSHOOT

$$\gamma_m > 5.100 \rightarrow \gamma_m > 15^\circ$$

• COMPUTE ω_c FROM SETTING TIME

$$T_s \leq 10; \omega_c \approx \frac{1}{T_s} \rightarrow \frac{1}{10} < 10 \rightarrow \omega_c > 10^{-1} \text{ Hz}$$

• CHOOSE A PROPER ω_c

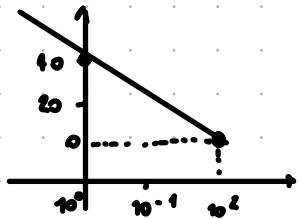
- CONSTRAINT FROM SETTING TIME $\rightarrow \omega_c > 10^{-1} \text{ Hz}$
- CONSTRAINT FROM NOISE $\rightarrow \omega_c < 40 \text{ KHz} \rightarrow \omega_c < 10^4 \text{ Hz}$
- CHOOSE A ω_c IN THE RANGE $\rightarrow \omega_c = 10^2 \text{ Hz}$

• TEST ω_c ON $R_1 \cdot G(s)$

$$R_1 G(\omega_c \cdot j) = \tan^{-1}(0/100) - \tan^{-1}(10^2/0) - \tan^{-1}(10 \cdot 10^2/1) - \tan^{-1}(10^2/100) \\ = 0 - (90^\circ) - (90^\circ) - (45^\circ) = -225^\circ$$

• CHECK IF THE RESULT RESPECT THE CONSTRAINT

- $\gamma_m = 180 - |R_1 G(\omega_c \cdot j)| = 180 - |-225| = -45^\circ$ UNSAT $\checkmark$ SOLUTION $\rightarrow$ CANCEL STABLE POLES OF $G(s)$
- $\gamma_m > 15^\circ$ $\leftarrow$ FROM OVERSHOOT



• CREATE R_2

- SUCH THAT $L \cdot R_1 \cdot R_2 \rightarrow$
- NUMERATOR OF R_2 IS A PART OF THE DENOMINATOR OF $G(s)$, SO THAT WE CAN CANCEL OUT SOME POLES
- DENOMINATOR OF R_2 IS $\pi(1+T \cdot s)$

$$G(s) = \frac{100}{s(10s+1)(s+100)} \\ R_2 = \frac{(10s+1)(s+100)}{\pi(1+Ts)} \quad R_1 = N \\ L = R_1 \cdot R_2 \cdot G(s) = \frac{100N}{s \pi(1+Ts)}$$

• BODE PLOT

- REMEMBER CONSTRAINTS ON ω_c
- CREATE THE BODE PLOT WITH THE ω_c WE CHOSED ($\omega_c = 10^2$)
- COUNT STEP BACK TO REACH 40° FROM $\omega_c \rightarrow$ TWO STEPS $10^0 \rightarrow 10^{-1} \rightarrow 10^{-2}$
- EACH STEP IS $+20 \text{ dB}$ ON γ -AXIS $\rightarrow 40 \text{ dB}$ TOTAL
- WE FOUND A POINT (10^0 ; STEP-20dB) $\rightarrow (10^0; 40 \text{ dB})$

• FIND N

- COMPUTE $L(0) \rightarrow L(s) = \frac{100N}{s \pi(1+Ts)} \rightarrow L(0) = 100N$ $\checkmark$ SKIP POLES AND ZERO THAT ARE EQUAL TO 0

$$\gamma \text{ VALUE IS } 20 \log_{10}(N) \rightarrow \text{FIND } N \rightarrow 40 = 20 \log_{10}(N) \rightarrow N = 100$$

$$L(0) = N \rightarrow 100N = 100 \rightarrow N = 1$$

• FIND T

- ONE STEP AFTER ω_c ON THE RIGHT
- $10^2 \rightarrow 10^3 \rightarrow T = 1/10^3$

• CREATE FINAL REGULATOR $R = R_1 R_2$

$$N = 1; T = 1/10^3 \\ R = \frac{(10s+1)(s+100)}{(1+s/10^3)}$$

- FEASIBLE $\#P \geq \#Z \checkmark$
- STABLE $\checkmark$

ADD A POLE THAT DOES NOT INTERFERE $\leftarrow$ SOLUTION NOT FEASIBLE $\checkmark$ $\leftarrow$

- WE KNOW: $10^2 < \omega_c < 10^4 \rightarrow \frac{1}{10^4} < T < 10$
- CHOOSE A T OUTSIDE THE RANGE AND AFTER ω_c

$$\rightarrow T < 1/10^4 \text{ CHOOSE } T = 1/10^5$$

$$R = \frac{(10s+1)(s+100)}{(1+s/10^3)(1+s/10^5)}$$

• FINAL CHECK ON CONSTRAINTS

$$L = R \cdot G(s) = \frac{(10s+1)(s+100)}{(1+s/10^3)(1+s/10^5)} \cdot \frac{100}{s(10s+1)(s+100)} = \frac{1}{s(1+s/10^3)(1+s/10^5)}$$

- L FEASIBLE $\checkmark \rightarrow \text{DECID} \geq \text{DECN}$
- $\gamma_m > 15^\circ$
- $\gamma_m = 180 - |L(\omega_c \cdot j)| \checkmark \text{ SAT.}$
- $T = \frac{1}{\omega_0} < 10 \text{ SEC} \checkmark \text{ SAT.}$

NOISE REQUIREMENT?

VFCN10 ES 3

SET CONTROLLER THAT UNDER DISTURBANCE (NOISE) CONDITION GUARANTEES...

$$G(s) = \frac{1}{s+10}$$

⑤ IDENTIFY THE SYSTEM

- P = POLES → NUMBERS THAT MAKES DENOMINATOR = 0
- Z = NUMBER THAT MAKE NUM. = 0
- # OF POLES → ORDER
- ORDER N = ORDER D → PROPER
- ORDER N ≠ ORDER D → STRICTLY PROPER
- STABLE → ALL POLES < 0
- UNSTABLE → ∃ P > 0

CONSTRAINTS

- NOISE HAVE NEGLECTIBLE VALUE IN FREQUENCY GREATER THAN 10 KHz
- OR: NOISE SIGNIFICANT COMPONENTS FOR FREQUENCY GREATER THAN 10 KHz
- OR: NOISE HAS AN ATTENUATION OF AT LEAST 20 dB
- $\epsilon_p = 0$
- OR: ERROR LOWER 5%
- $30 < T_{d2} < 300$
- OR: SETTLING TIME AT MOST 10 SECONDS
- NO OVERSHOOT
- OR: MAX $\xi = 0.2$

① NOISE

- 10 KHz = $10 \cdot 10^3 \text{ Hz} = 10^4 \text{ Hz}$
- BANDWIDTH $B < 10^5 \text{ Hz}$

WHAT IF "NOISE HAVE AN ATTENUATION OF AT LEAST 20 dB"?

CHOOSE A SUITABLE POWER OF S SUCH THAT $\frac{N}{S} G(s)$ HAVE GAIN $G = 0$

② ERROR:

- $\epsilon_m = 0$ MEANS $\eta = 1$
- $|N \cdot 1| < 5\%$ → $-5\% < N \cdot 1 < 5\%$ → $0.95 < N < 1.05$ → $N = 1$

③ SETTLING TIME

- $30 < T_{d2} < 300$ → $30 < 5T < 300$ → $10 < T < 100$
- $\frac{1}{100} < \omega < \frac{1}{10}$ (FLIP ω : $T = 1/\omega$)

• CHOOSE A CUT-OFF FREQUENCY:

- SMALLER IS BETTER
- $\omega_c = 1/100 = 10^{-2}$

THE CHOICE OF ω_c DEPENDS ON:

- SETTLING TIME (IT SET SOME BOUNDARIES)
- NOISE FREQUENCY (IT SET A BOUNDARY)

④ OVERSHOOT AND COMPUTE η_m REQUIREMENT

- IF NO OVERSHOOT → $\eta_m > 90^\circ$
- ELSE: $\xi = \frac{\eta_m}{100}$ → $\eta_m > 100 \xi$ → $\eta_m > 100 \cdot 0.2 \rightarrow \eta_m > 20^\circ$

④ CHECK IF CHOOSSED ω_c IS GOOD

- $\Delta G(\omega_c) \rightarrow \text{SUBSTITUTE } s \text{ WITH } \omega_c \cdot j$
- $\Delta G(\omega_c) \rightarrow \text{ARG}(1) - \text{ARG}(10^2 j + 10) = \tan^{-1}(0/1) - \tan^{-1}(10^2/10)$
- CHECK IF $\eta_m = 180 - |\Delta G(\omega_c)|$ IS GREATER THAN THE VALUE COMPUTE AT ④
- YES: $\rightarrow$
- NO → PICK A NEW ω_c CANDIDATE ω_c
- SIMPLIFY *

⑤ BODE PLOT

- X-AXIS: POWER OF 10
- "x" FOR POLES
- "o" FOR ZEROS
- LOCATE $(\omega_c; 0) \rightarrow \text{EX: } (10^{-2}; 0)$

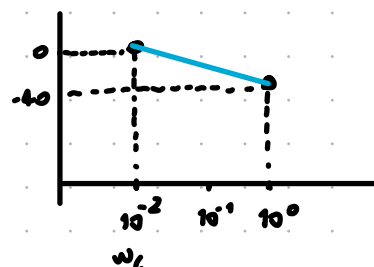
- COUNT THE STEP TO REACH $x=10^0$ FROM ω_c
- EACH STEP TO THE RIGHT → DECREASE γ OF 20 dB
- EACH STEP TO THE LEFT → INCREASE γ OF 20 dB
- EX: $10^{-2} \rightarrow 10^0$ 2 STEP → -40 dB
- CONNECT THE TWO LINE

⑥ FIND N

- THIS LOCATION YOU FOUND IS $(10^0; 20 \log_{10}(N))$
- YOU CAN FIND N ON THE POINT

ALTERNATELY TO FIND N:

$$N = G(0) \triangleq \text{NEED TO EXCLUDE THE ZEROS IN P AND Z}$$



⑦ PLOT THE FUNCTION

- MEET A POLE "x" → DECREASE m
- MEET A ZERO "o" → INCREASE m

- IF $\eta = \# \text{ZEROS IN P} - \# \text{ZEROS IN Z} = 0$; START → HORIZONTAL
- IF $\eta = 1$ →
- IF $\eta = -1$ →

8) CREATE C

- $C = \frac{N}{s^q} G(s)$
- $q = \# \text{ZEROS in } P - \# \text{ZEROS in } Z$

9) FINAL CHECK ON C

- STABLE $\rightarrow$ ALL POLES ARE NEGATIVE $\checkmark$
- FEASIBLE $\rightarrow \text{DEG}(N) \leq \text{DEG}(D)$ $\checkmark$

7) SIMPLIFY

- WE NEED ONE POLE BEFORE ω_c
- WE NEED ALL THE OTHER POLES BEFORE ω_c
- BY REMOVING A POLE WE INCREASE THE PHASE

EXAMPLE

$$C = \frac{N}{s^q} \cdot G(s) = \frac{100 s N \cdot \cancel{(s+200)}}{s^2 \cancel{(s+200)}} = \frac{100 N}{s}$$

$\uparrow$ zero
Signo

$\downarrow$ CUT OFF FREQUENCY

• FIND N

• $\omega_c = C$! ?

• FINAL CHECK ON C

- STABLE $\rightarrow$ ALL POLES ARE NEGATIVE $\checkmark$
- FEASIBLE $\rightarrow \text{DEG}(N) \leq \text{DEG}(D)$ $\checkmark$

